PHANTOM: Physics-Informed Hyperspectral Adversarial Network for Transformer-Optimized Methane Detection

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***Abstract*—Accurate detection of methane plumes in hyperspectral imagery is critical for environmental monitoring and greenhouse gas emission assessment. Traditional detection methods often struggle with complex atmospheric conditions, background variability, and subtle spectral signatures. We introduce the physics-informed hyperspectral adversarial network for transformer-optimized methane (PHANTOM) detection, a novel hybrid deep learning architecture that integrates physics-based constraints with advanced neural networks for robust methane plume segmentation. Our model's core is a U-Net generator with an integrated Transformer bottleneck, designed to capture both local and global spatial dependencies in the data to produce high-fidelity methane abundance maps. The framework is uniquely constrained by two physical principles: 1) a Beer–Lambert reconstruction loss, which models the physical attenuation of surface radiance through the gas plume using a dedicated background estimation network and 2) a reaction–diffusion partial differential equation (PDE) loss, which regularizes the solution to encourage the spatial structure of the predicted plume to be physically plausible. An adversarial discriminator further refines the output, ensuring the generated maps are structurally indistinguishable from ground truth labels. By deeply integrating these physical priors into the training objective, our method produces results that are more physically consistent, robust, and interpretable than traditional data-driven approaches. Benchmarked against the state-of-the-art HyperSTARCOP model, PHANTOM demonstrates superior performance across all evaluation metrics with seven times fewer parameters. It also transfers to the independent spaceborne EMIT dataset without architectural changes, roughly doubling the $F1$ score of matched filter (MF) baselines.**

***Index Terms*—Adversarial training, airborne visible/infrared imaging spectrometer–next generation (AVIRIS-NG), Beer–Lambert law, hyperspectral imaging, methane detection, physics-informed ML, segmentation, short-wave infrared (SWIR) spectroscopy, transformer.**

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I. Introduction

METHANE (CH_4) is a greenhouse gas whose 100-year global warming potential exceeds that of carbon dioxide by more than a factor of 25; combined with its much shorter atmospheric lifetime, this makes rapid and accurate detection critical for climate monitoring and mitigation. Accurately detecting and measuring methane emissions is crucial for managing both human-caused sources (landfills, agriculture, and fossil fuels) and natural ones (wetlands, geological seepage, and permafrost). This quantification is vital for pinpointing the main sources of emission and supporting mitigation efforts, providing a pathway to reduce CH_4 emissions, as pledged by nations under climate mitigation agreements [1].

Methane detection methodologies operate across multiple geographic scales, from localized ground-level analysis to expansive remote sensing. Ground-based methods include fixed-site measurements from high tower networks [2], rooftops [3], and other stationary platforms. For more dynamic, localized surveys, mobile sensing methods are employed using platforms like uncrewed ground vehicles (UGVs) [4], [5] and uncrewed aerial systems (UASs) [6], [7]. At the largest scales, methane plumes are detected using remote sensing methods that process hyperspectral data from aircraft and satellite sensors. Airborne imaging spectrometers such as airborne visible/infrared imaging spectrometer-next generation (AVIRIS-NG) [8] and the global airborne observatory (GAO) [9], and satellite sensors such as GHGSat [10], the earth surface mineral dust source investigation (EMIT) [11], precursore iperspettrale della missione applicativa (PRISMA), the environmental mapping and analysis program (EnMAP), and Carbon Mapper, have been used for methane plume detection. STARCOP [12], MethaneMapper [13], and MethaNet [14] are among the approaches applied to airborne hyperspectral datasets for methane plume detection. CH4net [15], the PRISMA deep-learning approach of Groshenry et al. [16], the multisatellite single-blind comparison by Sherwin et al. [17], and the satellite methane observations synthesis by Jacob et al. [18] constitute related work on identifying plumes from satellite-based datasets. Satellite-based methods have been comprehensively reviewed by Jiang et al. [19].

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In airborne and satellite hyperspectral imagery, methane absorption features lie in the short-wave infrared (SWIR) region of the electromagnetic spectrum, primarily around 1.65 and 2.3–2.4 μm. Several spectral methods have been developed to detect methane plumes, including column matched filters (MFs) [20], iterative maximum a posteriori differential optical absorption spectroscopy (IMAP-DOAS) and hybrid singular value decomposition (SVD) approaches [21], [22], and weighting function modified DOAS (WFM-DOAS) [23], to name a few. DOAS-based retrievals are accurate but computationally expensive, and the faster SVD approach is sensitive to surface features that resemble methane [21]. Among these methods, the column MF is the most widely adopted for plume detection, primarily due to its low computational cost, robustness across sensors, and a proven track record. However, it can also capture background artifacts along with the plume, potentially leading to inaccurate methane detection and often necessitating a human-in-the-loop to manually identify emission sources. Several spectral–spatial methods have been developed that combine traditional spectral techniques with deep-learning-based data-driven approaches to learn methane plumes from hyperspectral images [12], [13], [14], [16]. However, their predictions are not constrained by physics, which limits their physical consistency.

Physics-informed machine learning uses prior physical knowledge to improve learning through the training data (observational bias), the network architecture (inductive bias), or the training loss (learning bias) [24]. Such approaches have recently been adopted for methane detection, and they differ mainly in where the physics enters. Joyce et al. [25] attenuate PRISMA radiance with weather research and forecasting large-eddy simulation (WRF-LES) simulated plumes using HITRAN cross sections, training a convolutional neural network (CNN) to detect plumes and estimate emission rates without wind data. Rouet-Leduc and Hulbert [26] trade fidelity for scale, embedding analytical Gaussian plumes into Sentinel-2 band 12 to train a vision transformer. Si et al. [27] inject large eddy simulation plumes into EnMAP radiance through a radiative transfer transmittance and train multitask networks for retrieval, segmentation, and emission rate estimation. Batchu et al. [28] inject Lagrangian puff plumes into EMIT radiance with radiative transfer and screen detections with a spectral fit after inference. In these four works, physics enters through the training data. Pérez-Carrasco et al. [29] apply it after inference, rejecting MethaneSAT detections with implausible branched shapes, while FLAME [30] builds it into the architecture as a fixed log-domain Beer–Lambert MF layer fed by a learned per-pixel background.

Although these methods bring physics into methane detection, gaps remain. First, most of them learn from simulated plumes, whose realism limits performance on real scenes [27], [28], and false positives remain an open challenge [26], [28]. Second, where physics acts on the model or its output, it is either fixed in the architecture [30] or applied after inference [28], [29]; none requires during training that the predicted plume, together with the background, reproduces the measured radiance. Third, none constrains the spatial structure of the predicted plume during training with the physics of gas dispersion. FLAME does use physics in its training loss, through an auxiliary term that aligns its score with an MF retrieval [30], but none of these methods penalizes its own predictions for violating a physical law.

We propose a physics-informed hyperspectral adversarial network for transformer-optimized methane (PHANTOM) detection, a physics-informed spectral–spatial framework for methane plume detection that addresses these gaps. Trained on real annotated plumes [12], PHANTOM applies the Beer–Lambert law in the forward direction: a background estimator and the predicted abundance map reconstruct the observed SWIR bands, and because the methane absorption spectrum is fixed, a prediction that does not explain the observed attenuation is penalized through the reconstruction error. A reaction–diffusion partial differential equation (PDE) regularizer penalizes departures from the diffusive structure of the annotated plume, and an adversarial discriminator encourages realistic plume shapes. PHANTOM thus uses all three biases: observational, through the MF input [31]; inductive, through the background estimator and the exponential Beer–Lambert form; and learning, through the reconstruction and PDE losses, with the adversarial loss as a learned shape prior. The framework builds on a U-Net-based generative adversarial network (GAN) [32] with a Transformer bottleneck that captures global spatial context. We compare PHANTOM with HyperSTARCOP [12] on the STARCOP dataset and train and evaluate it on the independent methane alert and response system (MARS)-Hyperspectral EMIT dataset [33].

II. Methods

In this section, we describe NASA's Jet Propulsion Laboratory instrument AVIRIS-NG, MFs, network architecture, physics-informed machine learning and physical constraints, and loss functions and optimization.

A. Airborne Visible/Infrared Imaging Spectrometer–Next Generation

The AVIRIS-NG, using a pushbroom spectral mapping system, measures the wavelength range from 380 to 2510 nm with 5-nm spectral resolution. Its ground sampling distance is approximately 3–8 m [8]. We use the STARCOP dataset [12], built from 300 AVIRIS-NG Level 1B hyperspectral images of the 2019 Permian Basin campaign and sampled into 512×512 pixel chips. We use five SWIR bands at 2004, 2109, 2310, 2350, and 2360 nm: methane absorbs strongly at 2310–2360 nm, while its absorption at 2004 and 2109 nm is negligible. STARCOP describes the conversion of the radiance values to top-of-atmosphere (TOA) reflectance only for its simulated multispectral bands (WorldView-3).

B. Matched Filters

A MF [34] is a standard signal processing technique used to detect a known signal, or target signature, within a noisy background. In hyperspectral remote sensing, the MF is employed as a linear detector to maximize the signal-to-noise ratio, effectively identifying pixels whose spectral characteristics match a target of interest while suppressing the response from the background clutter. The MF response of pixel i is computed

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from its radiance vector $L_i \in \mathbb{R}^n$ over n bands, and the background mean $\mu \in \mathbb{R}^n$, both in μW·cm^{-2}·sr^{-1}·nm^{-1} (power per unit area, per unit solid angle, per unit wavelength), and the background covariance $C \in \mathbb{R}^{n\times n}$

$$\hat{z}_i = \frac{(L_i - \mu)^T C^{-1}(t(\mu))}{(t(\mu))^T C^{-1}(t(\mu))} \tag{1}$$

where $\hat{z}_i$ is the methane enhancement in ppm·m and $t(\mu) \in \mathbb{R}^n$ is the target spectrum, the change in radiance per unit enhancement. The numerator measures the background-suppressed correlation between the pixel's spectrum and the target signature, while the denominator normalizes the output. Mag1c [31] enhances the traditional MF by incorporating a sparsity prior, which assumes methane plumes are spatially rare, and an albedo correction to improve retrievals over surfaces with varying reflectance. However, despite these advancements, the mag1c output is not entirely free from background noise and spectral artifacts that can result in false positives. We provide the mag1c matched-filter output as an auxiliary input alongside SWIR data so the physics-informed model can jointly learn from measurements and imperfect matched-filter cues to segregate plume emissions from background clutter and suppress false positives. The MF input is the weighted mag1c of STARCOP [12], divided by 400 ppm·m and clipped to $[0.1, 1]$, then min–max normalized per tile to $[0, 1]$ and scaled by a factor of 0.2 to reduce the model's reliance on this preprocessed cue and encourage robust feature learning directly from the SWIR data.

C. Network Architecture

The PHANTOM framework is composed of three interconnected deep-learning blocks (neural networks) that are trained synergistically: a generator (G) to produce the methane abundance map, a discriminator (D) to ensure the map's structural realism, and a background estimator (B_{est}) to predict the background from the input SWIR image. The generated abundance map and estimated background facilitate the physics-informed reconstruction using the Beer–Lambert law. Details of each block are described below.

1) Generator Network (G): The core of our framework is a U-Net-based generator [35], designed to produce a high-fidelity methane abundance map. It takes a six-channel (SWIR + mag1c), 128 × 128 input tensor and progressively down-samples it through its encoder path to a 64 × 64 × 64 feature map at the bottleneck. To enhance its ability to model complex spatial relationships, this feature map is processed by a Transformer encoder [36], which captures global context without altering the tensor's dimensions. The transformer encoder comprises four layers, eight self-attention heads, and a feed-forward network dimension of 1024. The decoder path then symmetrically upsamples the features back to the original 128 × 128 resolution, using skip connections to preserve fine-grained details. A final 1 × 1 convolution produces a single-channel logit map, $\mathcal{L}$, which is passed through a sigmoid function to yield the final plume probability abundance map $\hat{A} \in (0, 1)^{128\times 128}$, with $\hat{A}_i$ its value at pixel i. The generator network is shown in Fig. 1.

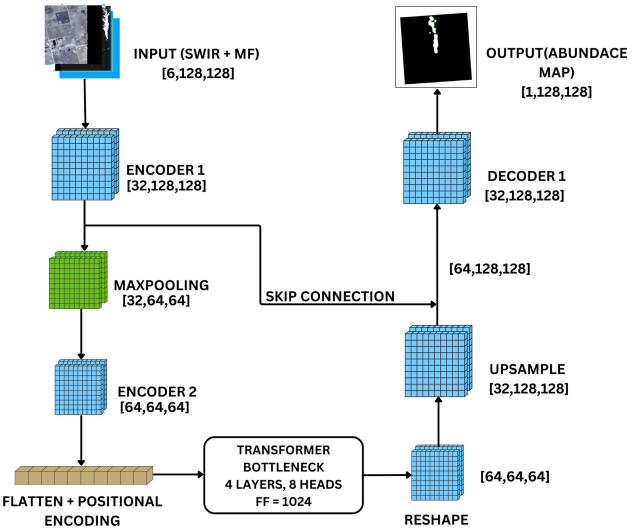


Fig. 1. U-Net generator with Transformer bottleneck.

2) Discriminator Network (D): The discriminator acts as a trainable loss function that guides the generator to produce methane abundance maps that are structurally indistinguishable from the ground truth labels [12] provided by STARCOP. As a critic, it is trained by comparing the "real" ground truth masks against the "fake" abundance maps from the generator. For the discriminator, we employ a PatchGAN architecture [37]. Unlike a traditional discriminator that outputs a single verdict for the entire image, the PatchGAN acts as a local critic rather than a global one. It takes a single-channel 128 × 128 input mask, real or fake, and downsamples it to a 31 × 31 feature map. Each element in this map corresponds to the discriminator's judgment on the realism of a specific, overlapping patch of the input. By evaluating the image at this granular level, the discriminator penalizes local inconsistencies and forces the generator to produce sharp, realistic high-frequency details, clean plume edges, and textures.

3) Background Estimator (B_{Est}): To accurately enforce the Beer–Lambert law, it is essential to disentangle the background radiance from the input SWIR signal attenuated by the methane plume. For this purpose, we introduce a dedicated background estimator block, B_{est}. This lightweight CNN takes the five-channel, 128 × 128 SWIR image as input and produces an estimated background $\hat{B} \in [0, 1]^{5\times 128\times 128}$ in the same normalized units as the input. By separating the background estimation, the model can more robustly apply physics-based learning. This creates a critical feedback loop: the background estimator provides a "clean slate" of the scene, and the generator's plume map is used with the Beer–Lambert law to reconstruct the original SWIR image. If the generated plume is physically inaccurate, the reconstruction will be poor, and the reconstruction loss penalizes the generator.

D. Physics-Informed Machine Learning

Physics-informed machine learning, as reviewed by Karniadakis et al. [24], integrates physical knowledge into learning through the training data, the network architecture, or the training loss. Physics-informed neural networks (PINNs), pioneered by Raissi et al. [38], represent a class of deep learn-

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ing models that integrate governing physical laws directly into the training process. Unlike traditional data-driven approaches, PINNs are constrained by these physical principles—typically expressed as PDEs—by incorporating the PDE residual as a component of the total loss function. This hybrid, physics-informed methodology makes them exceptionally effective for solving both forward and inverse problems, particularly in domains with complex geometries or high dimensionality where traditional numerical solvers may falter. As a result, PINNs have become a cornerstone of scientific machine learning (SciML), with notable applications in fields such as high-speed aerodynamics, incompressible fluid flow, heat transfer, and phase-change dynamics. Building on this idea, PHANTOM adds physical laws to its training loss, but its network maps an image to a plume map rather than approximating the solution of a PDE.

In our work, we apply physics-informed machine learning to hyperspectral remote sensing for methane detection. We identified two core physical principles that govern the appearance and behavior of methane plumes in aerial imagery. The first is the principle of radiative transfer [39], described by the Beer–Lambert law, which models how surface radiance is attenuated as it passes through the gas. The second is the principle of gas dispersion [40], which we approximate using a reaction–diffusion equation to enforce a physically plausible spatial structure on the plume. These two principles are translated into differentiable loss terms that constrain the network's objective function, ensuring the final predictions are not only data-driven but also physically coherent. Implementation details on both of these physical principles in PHANTOM are elaborated below.

1) Beer–Lambert Reconstruction: The relationship between the observed radiance, the background radiance, and the gas concentration is governed by the Beer–Lambert law [41]. This law states that the observed radiance is the result of the background radiance being exponentially attenuated by the target material. We model this relationship to reconstruct the original SWIR image

$$\hat{X}_{SWIR} = \hat{B} \odot \exp\left(-\hat{A} \odot \boldsymbol{\kappa}\right) \tag{2}$$

where $\hat{X}_{SWIR}$ is the reconstructed SWIR image, $\hat{B}$ is the estimated background, $\hat{A}$ is the predicted methane abundance map, $\boldsymbol{\kappa} \in \mathbb{R}^5$ is the vector of methane absorption coefficients for the five bands derived from EMIT data resources [11]—a valid approach as methane's absorption spectrum is a sensor-independent physical constant—and $\odot$ denotes elementwise multiplication.

2) Reaction–Diffusion PDE Regularization: To impose a physically plausible spatial structure on the predicted plume, we introduce a physics-based regularizer based on a reaction–diffusion equation [40]. This soft constraint guides the learning process by penalizing solutions that are not spatially smooth or do not conform to the natural dynamics of gas dispersion. The regularizer is formulated using the residual of the PDE operator, $\mathcal{R}$, which is applied to a concentration map U

$$\mathcal{R}(U) = D_c \nabla^2 U - \lambda_c U \tag{3}$$

where $U \in [0,1]^{128\times128}$ is a map on the pixel grid [$\hat{A}$ or Y in (7)], ∇^2 is the discrete Laplacian, which enforces smoothness, and D_c and λ_c are the diffusion and reaction (decay) coefficients.

By penalizing the difference between the PDE residuals of the predicted map and the annotated plume (7), the regularizer encourages the predicted plume to vary smoothly in space, as expected of a diffusing gas, rather than to form abrupt, speckled patches. It therefore shapes the spatial structure of the predicted plume, as the ablation in Section IV-C shows. The term is a spatial prior on a single image, with lumped coefficients in pixel units, rather than a transport model; it omits advection because the STARCOP dataset provides no local wind data, and an advection variant is tested on EMIT in Section IV-E.

E. Loss Functions and Optimization

The parameters of the generator (G), discriminator (D), and background estimator (B_{est}) are optimized using a composite objective function that synergistically combines data-driven, adversarial, and physics-informed losses. The generator and background estimator are trained jointly to minimize a total objective, while the discriminator is trained in an alternating fashion to distinguish real from generated plume maps.

1) Adversarial and Segmentation Losses: The training process is anchored by an adversarial dynamic between the generator and the discriminator. The discriminator D learns to distinguish between real ground truth labels Y and the synthetic abundance maps $\hat{A}$ produced by the generator. In response, the generator G is trained to produce maps that are indistinguishable from real data, thereby minimizing the adversarial loss $\mathcal{L}_{adv}$

$$\mathcal{L}_{adv} = \mathcal{L}_{BCE}\left(D\left(\hat{A}\right), \mathbf{1}\right) \tag{4}$$

where $\mathbf{1}$ is a 31×31 tensor of ones, matching the PatchGAN output. To ensure pixel-level accuracy, the generator's output is also supervised by a hybrid segmentation loss, $\mathcal{L}_{seg}$. This loss is a direct sum of a binary cross-entropy (BCE) term and a Dice loss term, promoting both local precision and global spatial overlap

$$\mathcal{L}_{seg} = -\frac{1}{N}\sum_{i=1}^{N}\left[Y_i \log\left(\hat{A}_i\right) + (1-Y_i)\log\left(1-\hat{A}_i\right)\right] + \left(1 - \frac{2\sum_{i=1}^{N}\hat{A}_i Y_i + \epsilon}{\sum_{i=1}^{N}\hat{A}_i + \sum_{i=1}^{N} Y_i + \epsilon}\right) \tag{5}$$

where $Y_i \in \{0,1\}$ is the ground-truth label of pixel i (1 = plume, 0 = background), $\hat{A}_i \in (0,1)$ is the prediction, $N = 128 \times 128$ is the number of pixels per tile, and ϵ is a small constant for numerical stability.

2) Physics-Informed Losses: The two physical constraints are translated into differentiable loss functions. The Beer–Lambert reconstruction loss ($\mathcal{L}_{recon}$) is formulated as the L1 norm, averaged over all bands and pixels, between the original SWIR image X_{SWIR} and the reconstructed image $\hat{X}_{SWIR}$ from (2), enforcing consistency with radiative transfer physics

$$\mathcal{L}_{recon} = \left|\left|X_{SWIR} - \hat{X}_{SWIR}\right|\right|_1. \tag{6}$$

The reaction–diffusion PDE regularization loss ($\mathcal{L}_{PDE}$) penalizes solutions that deviate from the expected behavior of gas

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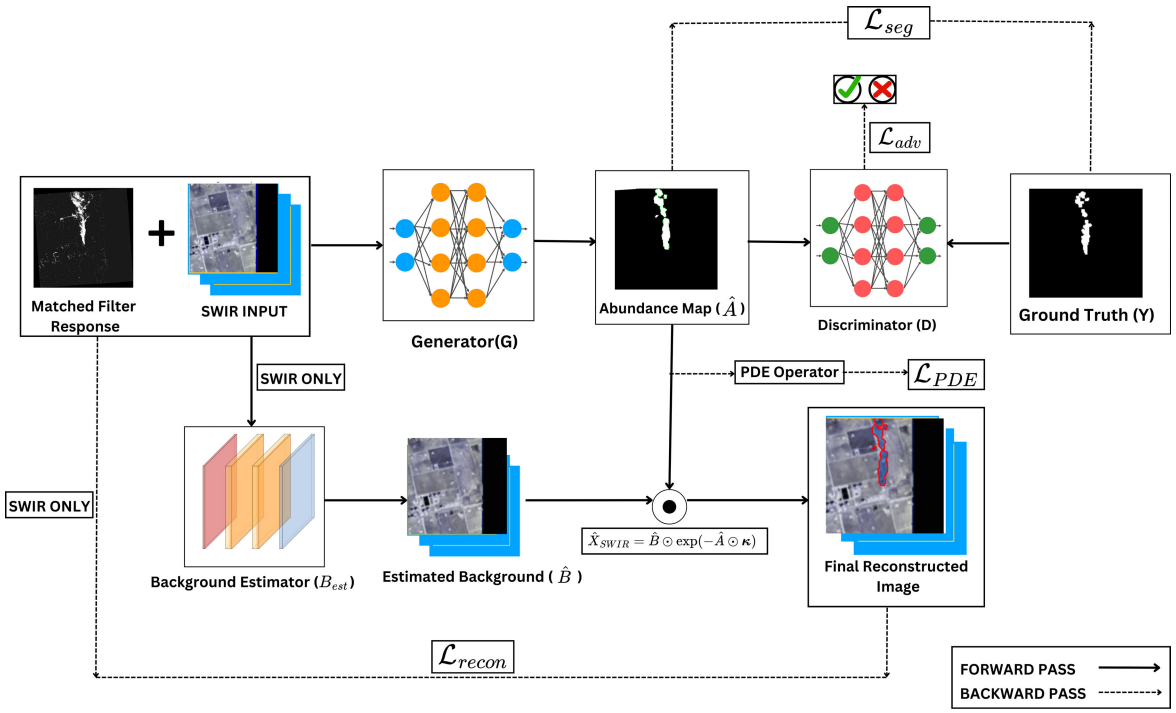


Fig. 2. Proposed PHANTOM architecture, illustrating the synergistic training of the generator (G), discriminator (D), and background estimator (B_{est}). The forward pass shows the data flow for generating the methane abundance map, while the backward pass indicates the flow of gradients from the data-driven, adversarial, and physics-informed loss functions that jointly optimize the framework.

dispersion. It is calculated as the L1 norm, averaged over all pixels, of the difference between the PDE residuals [from (3)] of the prediction $\hat{A}$ and the ground truth Y, masked to apply only within the true plume area

$$\mathcal{L}_{PDE} = \left|\left|\left(\mathcal{R}\left(\hat{A}\right) - \mathcal{R}\left(Y\right)\right) \odot Y\right|\right|_1. \tag{7}$$

3) Total Objective Function: The complete objective for the generator and background estimator is a weighted sum of these individual components, balancing their respective contributions

$$\mathcal{L}_{G_{total}} = \lambda_{seg}\mathcal{L}_{seg} + \lambda_{adv}\mathcal{L}_{adv} + \lambda_{recon}\mathcal{L}_{recon} + \lambda_{pde}\mathcal{L}_{PDE}. \tag{8}$$

The network architecture with both forward and backward passes is shown in Fig. 2.

III. Experiments

To validate the performance of the proposed PHANTOM framework, we conducted a series of experiments using AVIRIS-NG datasets [12] and the independent EMIT dataset MARS-Hyperspectral [33]. This section details the dataset preprocessing, our specific implementation choices, the training procedure, and the metrics used for evaluation.

A. Datasets and Preprocessing

All input data tiles (SWIR bands, MF, and ground truth masks) were resized to a uniform spatial dimension of 128×128 pixels using bilinear resampling. Masks were then binarized ($Y_i = 1$ where the resampled value is > 0). All inputs are dimensionless after normalization. To ensure numerical stability during training, each of the five SWIR bands and the MF channel were normalized independently to the range $[0, 1]$. We use the original STARCOP [12] split: 3425 training tiles and 342 test tiles.

To evaluate PHANTOM on an independent sensor, we use MARS-Hyperspectral [33], built from the spaceborne EMIT imaging spectrometer on the International Space Station, with 60 m pixels and plumes annotated by human analysts. Each capture is a 256×256-pixel tile of one site from a single EMIT overpass. The captures are split by acquisition date into 5746 training captures, mainly from 2022 to 2023, and 2316 validation and 1708 test captures, mainly from 2024; the test set also includes 200 full scenes containing 311 plumes. MARS provides the plume labels and three MF products [mag1c, an MF, and a wide-window MF (WMF)], but not the SWIR bands. We therefore took the same five SWIR bands from the EMIT level 1B (L1B) at-sensor radiance product [42] and placed them on the MARS grid. The sixth input channel is a MARS MF product, mag1c or WMF, transformed, min–max normalized per tile, and scaled by 0.2 exactly as for AVIRIS-NG; a third variant also receives the three red, green, blue (RGB) bands as extra inputs (see Section IV-D).

B. Implementation Details

The entire framework was implemented using the PyTorch library on an Intel Core i9-14900K processor with 64 GB of system memory. GPU acceleration was provided by a single NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. The key architectural and training parameters are outlined below:

1) Training Hyperparameters: The framework was trained for 40 epochs with a batch size of 8. All three networks (G, D, and B_{est}) were optimized using the Adam optimizer

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TABLE I

COMPARISON OF METHANE DETECTION PERFORMANCE. OUR PROPOSED PHANTOM MODEL IS BENCHMARKED AGAINST THE BASELINE AND HYPERSTARCOP MODELS. ALL METRIC VALUES ARE IN PERCENT (%); PARAMS IS THE NUMBER OF PARAMETERS OF THE NETWORK USED AT INFERENCE. PHANTOM IS REPORTED BOTH AS THE RELEASED CHECKPOINT AND AS MEAN ± STANDARD DEVIATION OVER TEN RANDOM SEEDS. THE ARROWS INDICATE WHETHER A HIGHER (↑) OR LOWER (↓) VALUE IS BETTER

Model	Params	F1 (strong) ↑	F1 (weak) ↑	FPR by tile ↓	AUPRC ↑
Baseline, mag1c + morpho.	–	67.45	39.95	75.43	N/A
HyperSTARCOP, only mag1c	6.63 M	74.15 ± 6.10	47.57 ± 4.17	52.11 ± 10.98	49.41 ± 5.49
HyperSTARCOP, mag1c + rgb	6.63 M	81.96 ± 3.71	43.42 ± 5.72	43.66 ± 7.36	51.99 ± 2.76
PHANTOM, mag1c + swir (released checkpoint)	**0.96 M**	**100.00**	**96.65**	11.43	62.42
PHANTOM, mag1c + swir (10 seeds)	**0.96 M**	98.07 ± 3.45	87.13 ± 8.61	$\mathbf{7.31 \pm 2.22}$	$\mathbf{63.44 \pm 3.13}$

with a learning rate of 3×10^{-5} and betas of $(0.5, 0.999)$. To accelerate training, we utilized automatic mixed precision (AMP), a technique that combines single-precision (FP32) and half-precision (FP16) data types to accelerate training and reduce memory usage without losing model accuracy. The generator, the only network used at inference, has 0.96M parameters; the discriminator (34 401 parameters) and the background estimator (3141 parameters) are used only during training. Training for 40 epochs on the 3425 STARCOP training tiles takes 10–11 min on the GPU.

2) Loss Function Weights: The total generator objective in (8) was balanced with the following weights: the segmentation loss weight λ_{seg} was set to 1.0 (for the combined BCE and Dice terms), the adversarial weight λ_{adv} was 0.05, the Beer–Lambert reconstruction weight λ_{recon} was 1.0, and the PDE regularization weight λ_{pde} was 0.5. This weighting scheme prioritizes the primary data-fitting objectives (segmentation and reconstruction) while leveraging the adversarial and PDE losses as regularizers to encourage structural realism and physical consistency.

3) Physics Coefficients: The PDE residual in (3) was parameterized with a diffusion coefficient $D_c = 0.5$ and a reaction (decay) coefficient $\lambda_c = 0.01$. These values were selected to impose a soft physical constraint, where the diffusion term primarily encourages a spatially smooth plume structure, and the smaller reaction term provides mild regularization against excessively high abundance values.

4) Training on EMIT: PHANTOM was initialized from the released AVIRIS-NG model and trained on the MARS training split for 60 epochs with the MARS tiling and sampling. The architecture, losses, loss weights, and optimizer are unchanged; only the absorption coefficients κ were recomputed for the EMIT bands. The checkpoint with the best validation $F1$ was kept, and training was repeated with five random seeds.

C. Evaluation Metrics

To ensure a fair and direct comparison with the HyperSTARCOP [12] baseline, we adopt its exact evaluation methodology. The overall pixel-level segmentation performance is first assessed using the area under the precision-recall curve (AUPRC), a threshold-independent metric well-suited for the class imbalance inherent in plume detection. To evaluate performance at the tile level, the continuous prediction maps are converted into binary detection decisions: a tile is classified as containing a plume if its segmentation mask, after applying a 0.5 threshold, contains more than ten active pixels. Using this rule, we measure the model's reliability against false alarms (FAs) by calculating the false positive rate (FPR) on the subset of test tiles known to be plume-free. Finally, to provide a nuanced view of detection capability, the plume capture rate is reported as an $F1$ score and stratified by emission strength, with separate evaluations for strong (>1000 kg/h) and weak (≤1000 kg/h) plume events. On EMIT, we use the evaluation code released with MARS [33] at its threshold of 0.5: $F1$ and AUPRC are computed over the 1708 test captures, and detected plumes and FAs over the 200 full scenes containing 311 plumes.

IV. RESULTS AND DISCUSSION

A. Quantitative Analysis

The performance of our proposed PHANTOM framework was benchmarked against the baseline MF and two variants of the HyperSTARCOP model, with the results for these methods reported as published in their original work [12]. The quantitative results, presented in Table I, demonstrate that PHANTOM achieves state-of-the-art performance on the STARCOP benchmark, substantially outperforming all other methods across every evaluation metric. Because two trainings of the same configuration can differ by several points of $F1$ (see Section IV-C), Table I reports PHANTOM both as the released checkpoint and as the mean and standard deviation over ten trainings with different random seeds. In both cases, PHANTOM outperforms all other methods on every metric.

In terms of plume capture rate, PHANTOM shows exceptional capability. The released checkpoint achieves a perfect $F1$ score of 100.00% for strong plumes, indicating it successfully detected all major emission events in the test set. More notably, its $F1$ score of 96.65% on weak plumes is more than double that of the best-performing HyperSTARCOP model (47.57%), highlighting its superior sensitivity to subtle and diffuse methane signatures that are often missed by other methods. This high sensitivity is further evidenced by the model's ability to successfully detect plumes with emission rates as low as 97.34 kg/h.

The model also demonstrates a significant improvement in reliability. With an FPR of only 11.43%, PHANTOM generates nearly four times fewer FAs than the next best model (43.66%). This drastic reduction in false positives is a critical advantage for practical monitoring applications, as it minimizes the need for costly manual verification of potential emission sites and increases confidence in automated detection systems. At the pixel level, the overall segmentation quality, as measured by the AUPRC, is also markedly higher at 62.42% compared to HyperSTARCOP's 51.99%, confirming that the

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improvements are not just in tile-level detection but also in the fidelity of the generated abundance maps.

PHANTOM also has seven times fewer parameters than HyperSTARCOP (0.96M against 6.63M). Per STARCOP tile, measured with the PyTorch FLOP counter for one forward pass, it requires 1.79 giga floating-point operations (GFLOPs) against 26.8 GFLOPs for HyperSTARCOP. Most of this 15-fold difference comes from the output resolution: PHANTOM predicts on a 128×128 grid, which has 16 times fewer pixels than the 512×512 grid of HyperSTARCOP, and at equal resolution the two models would require a similar number of operations. Thus, PHANTOM has a smaller model size, not a lower computational cost at equal resolution.

The component ablation in Section IV-C shows how each part of the framework contributes to this performance. The Transformer bottleneck contributes most: it captures the global spatial context that confirms locally detected candidates, as the gradient-weighted class activation mapping (Grad-CAM) analysis below illustrates, and the adversarial loss further improves the detection of weak plumes. The physics-informed terms complement these components by constraining the solution itself: the Beer–Lambert reconstruction loss ties each prediction to the measured SWIR radiance and makes training more consistent, and the reaction–diffusion regularizer yields smoother, more diffusion-like plume maps. PHANTOM also differs from HyperSTARCOP in its input: while HyperSTARCOP relies on RGB visual context, PHANTOM learns directly from the SWIR bands that contain methane's spectral absorption signature, although the effect of this choice was not isolated in our experiments.

B. Explainable AI (XAI) Analysis

To unlock the black box of the PHANTOM framework and interpret its decision-making process, we applied Grad-CAM [43], as shown in Fig. 3. The analysis reveals a sophisticated two-stage reasoning process that explains the model's high performance: a local feature detection stage followed by a global context verification stage. Grad-CAM was applied to the final U-Net encoder block to visualize local features (fourth column) and after the Transformer bottleneck to visualize global context (fifth column).

C. Component Ablation

Each component from PHANTOM was removed on its own, with the rest of the training unchanged: −Transformer replaces the Transformer bottleneck with two 3×3 convolutional layers of the same width; −GAN removes the discriminator ($\lambda_{adv} = 0$); −Beer–Lambert removes the reconstruction loss and the background estimator ($\lambda_{recon} = 0$); and −PDE removes the PDE term ($\lambda_{pde} = 0$). Every variant was trained on AVIRIS-NG with the same three random seeds. Table II gives the results.

The Transformer bottleneck is the most important component: removing it lowers AUPRC from 62.9 to 28.6 and $F1$ (weak) from 88.0% to 5.3%, and in one of the three seeds no test pixel exceeds the threshold. Without the global context that confirms locally detected candidates, as in the Grad-CAM analysis above, the network predicts background almost everywhere, which also explains its low FPR by tile. Removing the adversarial loss lowers AUPRC by 3.6 and pixel $F1$ by 3.4 points; the drop is largest for weak plumes (9.9 points of $F1$, against 3.5 for strong plumes), where the pixelwise loss alone gives the least signal. Removing the Beer–Lambert loss changes the mean scores little, but the pixel $F1$ standard deviation rises from 2.40 to 8.22: by requiring the predicted plume and background to reproduce the measured radiance, the loss ties the prediction to the observed methane absorption and narrows the range of solutions that different training runs reach.

TABLE II

COMPONENT ABLATION ON AVIRIS-NG. MEAN (TOP) AND STANDARD DEVIATION (BELOW) IN PERCENT OVER THE SAME THREE SEEDS FOR EVERY VARIANT. PIXEL $F1$ IS COMPUTED OVER ALL TEST PIXELS AT THRESHOLD 0.5; $F1$ (STRONG), $F1$ (WEAK), AND FPR BY TILE FOLLOW THE TILE RULE OF TABLE I. * LOW ONLY BECAUSE THIS VARIANT DETECTS ALMOST NOTHING. BEST MEAN PER COLUMN IN BOLD

Variant	AUPRC ↑	Pixel F1 ↑	F1 (strong) ↑	F1 (weak) ↑	FPR by tile ↓
Full model	**62.87**	56.85	**98.25**	87.96	9.14
	±1.07	±2.40	±3.04	±6.07	±1.71
−PDE	61.48	**60.01**	**98.25**	**89.20**	8.38
	±4.52	±1.58	±3.04	±7.71	±3.15
−Beer–Lambert	62.23	58.67	97.66	84.26	6.29
	±3.54	±8.22	±2.68	±5.78	±1.51
−GAN	59.29	53.42	94.74	78.09	7.62
	±2.70	±4.61	±5.26	±12.64	±2.01
−Transformer	28.61	23.08	16.37	5.25	**3.81***
	±3.59	±20.05	±14.18	±4.57	±3.30

The PDE term targets the spatial structure of the map rather than detection. At $\lambda_{pde} = 0.5$ it makes up only 0.2% of the training loss, and removing it changes neither AUPRC nor pixel $F1$ (ten seeds each, Welch $p = 0.49$ and 0.46). Retrained with $\lambda_{pde} = 36$ (13% of the training loss) on the same three seeds, the term smooths the predicted maps. The mean roughness $|\nabla^2 \hat{A}|$ inside annotated plumes, four times the departure of the predicted probability from the average of its four neighbors, falls from 0.42 without the term to 0.33 at $\lambda_{pde} = 0.5$ and 0.26 at $\lambda_{pde} = 36$ (paired $p = 0.007$ for 36 against 0.5), while detection is unchanged (AUPRC 59.64 ± 2.72, pixel $F1$ 57.33 ± 3.02). Fig. 4 shows that the probability then decreases gradually from the plume core to the background, as expected of a diffusing gas. The smoothing also applies to false detections, such as the patch to the right of the 52 px plume, so the term improves the spatial coherence of the maps rather than detection accuracy.

D. Generalization to EMIT

Table III evaluates PHANTOM on MARS-Hyperspectral, acquired from space at a spatial resolution about ten times coarser than AVIRIS-NG, with no change to the architecture. With the WMF input, PHANTOM roughly doubles the $F1$ score of every MF baseline and raises three to nine times fewer FAs, while detecting a similar number of plumes. The MF product has the largest effect for both networks: replacing mag1c with WMF raises the $F1$ of PHANTOM from 28.00 to 51.61 and that of the U-Net from 43.59 to 63.07. PHANTOM with WMF reaches a higher $F1$ and detects more plumes than the U-Net with RGB and mag1c, although with a lower AUPRC and more FAs. With the same input product, the U-Net is better: with WMF it reaches an AUPRC of

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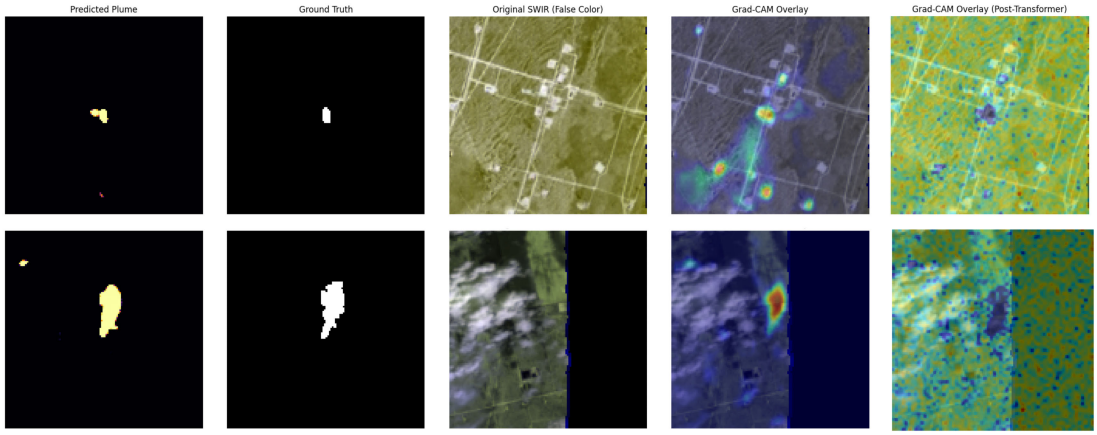


Fig. 3. To interpret PHANTOM with explainable AI (XAI), we applied Grad-CAM [43] to the final U-Net encoder block (fourth column of Fig. 3) and after the Transformer bottleneck (fifth column). The heatmaps reveal a two-stage process: local detection followed by global verification. In the cluttered scene of the top row, the encoder places a strong hotspot on the plume but also flags roads; the Transformer, which weighs features across the whole scene, dismisses the road artifacts while keeping the plume. In the bottom row, the encoder activation stays on the plume and does not spread into the adjacent cloud of similar shape, suggesting that it responds to the methane signature rather than to shape alone, and the Transformer confirms this detection. These findings suggest that PHANTOM's performance rests on local detection followed by global verification.

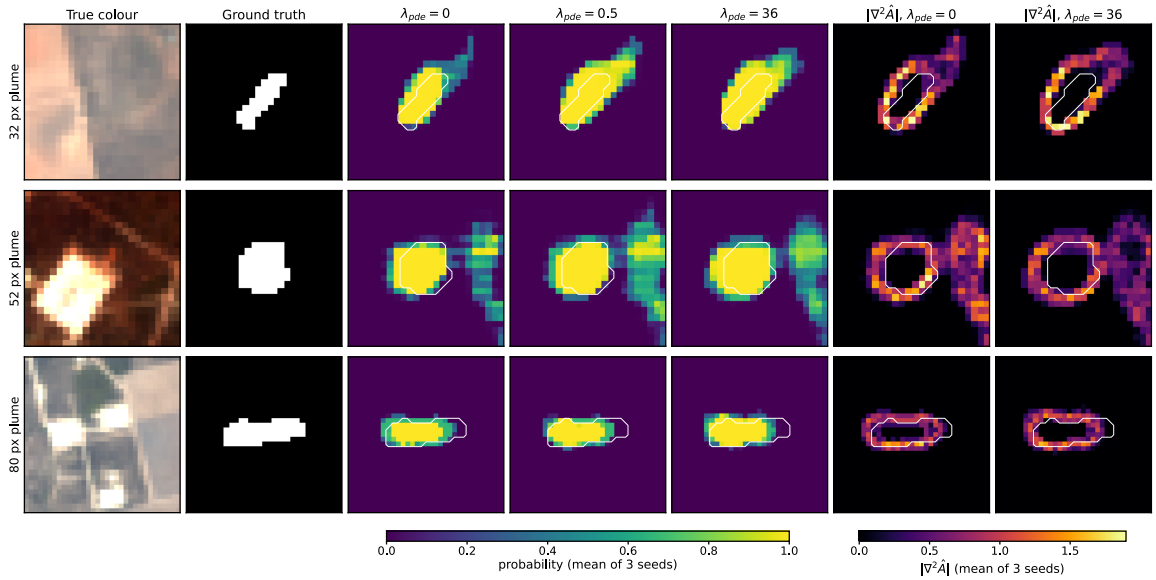


Fig. 4. Effect of the PDE weight on the predicted maps, on AVIRIS-NG test plumes at the 20th, 40th, and 60th percentile of plume size (cropped around the plume; white contour: annotation). Probability maps for $\lambda_{pde} = 0$, 0.5, and 36, and the roughness $|\nabla^2\hat{A}|$ for 0 and 36, each averaged over three seeds. Higher weight gives smoother maps, mostly at the plume boundary.

68.26 against 35.38 and raises about a third as many FAs. PHANTOM reaches 92% of its detections (195 against 213) with one seventh of its parameters (0.96M against 6.69M), but requires about 24% more computation per EMIT scene (483 against 391 GFLOPs), with both models working at the native 60 m resolution.

Part of this gap is attributable to missing surface context. The U-Nets combine RGB reflectance with the MF product [33], so they can use the visible appearance of the surface to reject MF artifacts, much as an analyst would [12]; PHANTOM receives only the SWIR bands and the MF, and cannot always separate a plume from an MF artifact over a bright surface. Adding the three RGB bands as extra inputs, with no other change, lowers FAs by 35% and raises precision from 56.3% to 63.4% at the same $F1$ (last row of Table III).

Fig. 5 shows PHANTOM on three test captures. The MARS reference plumes come from the public NASA EMIT level 2B (L2B) plume complexes [44] for the same acquisitions; NASA delineates plumes differently from the MARS analysts, so the intersection over union (IoU) values in the figure are not comparable with Table III. In all three captures, PHANTOM recovers the plume core and misses mainly its diffuse outer part; its false positives are small isolated patches, and it does not respond to the along-track striping in the mag1c map of the 75th-percentile capture.

E. Limitations

The reaction–diffusion term assumes that methane spreads equally in all directions with a single constant coefficient. It therefore ignores transport by the wind, which elongates a plume downwind, and local turbulence, which makes real plumes meander, break into puffs, or spread unevenly.

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TABLE III

GENERALIZATION TO MARS-HYPERSPECTRAL (EMIT). $F1$ AND AUPRC IN PERCENT OVER THE 1708 TEST CAPTURES (1704 FOR MAG1C INPUTS); DETECTED PLUMES AND FAS OVER 200 FULL SCENES CONTAINING 311 PLUMES. MF AND U-NET VALUES ARE AS PUBLISHED IN [33]; PHANTOM VALUES ARE MEAN ± STANDARD DEVIATION OVER FIVE SEEDS. VARIANTS WITH WIND AND LOCATION INPUTS ARE OMITTED, AS PHANTOM RECEIVES NEITHER. †ADDED AFTER THE OTHER EXPERIMENTS TO TEST THE EFFECT OF RGB CONTEXT.–: NOT REPORTED FOR MFS. BEST VALUE PER COLUMN IN BOLD

Model	F1 ↑	AUPRC ↑	Detected ↑	FA ↓
MF, mag1c	22.45	–	187	41,903
MF, mf	27.70	–	217	110,383
MF, wmf	23.40	–	**225**	79,273
U-Net, rgb + mag1c	43.59 ± 3.17	40.87 ± 2.75	137 ± 21	$5{,}789 \pm 2{,}157$
U-Net, rgb + wmf	$\mathbf{63.07 \pm 2.46}$	$\mathbf{68.26 \pm 3.06}$	213 ± 20	$\mathbf{3{,}565 \pm 1{,}830}$
PHANTOM, mag1c + swir	28.00 ± 1.60	17.62 ± 1.58	128 ± 30	$12{,}678 \pm 7{,}551$
PHANTOM, wmf + swir	51.61 ± 0.85	35.38 ± 0.78	195 ± 13	$12{,}671 \pm 2{,}654$
PHANTOM, wmf + swir + rgb†	51.99 ± 0.63	37.62 ± 1.27	191 ± 23	$8{,}236 \pm 3{,}050$

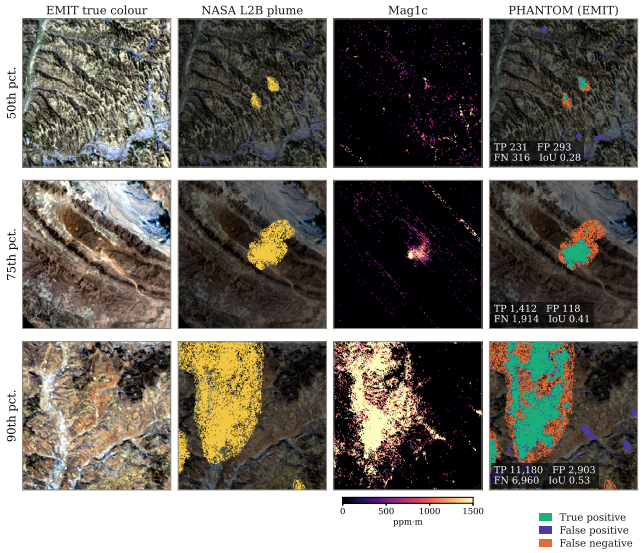


Fig. 5. PHANTOM trained on EMIT, applied to three MARS test captures at the 50th, 75th, and 90th percentile of IoU among the 85 test captures that contain a NASA plume and lie away from the swath edge. Columns: true color from EMIT L1B [42], NASA L2B reference plume [44], mag1c [31], and the PHANTOM prediction at threshold 0.5 (true positives, false positives, and false negatives).

To test advection, we added $-A\,(\mathbf{u}\cdot\nabla U)$ to (3), with $\mathbf{u}$ the goddard earth observing system, forward processing (GEOS-FP) 10 m wind vector provided with each MARS capture [33], and trained PHANTOM on EMIT with weights fixed before training ($\lambda_{pde} = 36$, $A = 11$). The advection term raised $F1$ from 52.31 to 54.88 (Welch $p = 0.006$, five seeds), but the same term with the wind vectors randomly permuted across training captures raised it just as much (55.20; $p = 0.54$), and AUPRC did not differ ($p \geq 0.25$): the gain comes from the advection term itself, not from the wind. The wind does agree with the annotated plumes, whose axes lie on average 25° from the wind direction, against 45° if unrelated, although analysts see the wind when annotating; adding it as an input to the MARS U-Net changed $F1$ by only 0.61 points, less than the standard deviation between training runs [33]. Wind is used mainly after detection to estimate emission rates over the plume mask [45], [46].

The PDE weight also sets the operating point. At $\lambda_{pde} = 0.5$ the term makes up only 0.2% of the training loss on AVIRIS-NG (see Section IV-C); on EMIT, raising it to 36 increased AUPRC from 35.38 to 41.30 ($p < 0.001$) without changing $F1$ ($p = 0.33$), with fewer detections (146 against 195) and fewer FAs (5376 against 12 671). Finally, on EMIT PHANTOM lacks the surface context of RGB imagery (see Section IV-D).

V. CONCLUSION

In conclusion, PHANTOM outperforms existing models on the STARCOP benchmark in both detection accuracy and reliability, particularly for weak plumes, with seven times fewer parameters than HyperSTARCOP. Its physics-informed terms tie each prediction to the measured radiance and to a diffusion-like spatial structure, which makes training more consistent and the predicted maps smoother. On the independent spaceborne EMIT dataset, PHANTOM transfers without architectural changes, roughly doubling the $F1$ score of MFs and reaching 92% of the detections of the U-Net with RGB and WMF input with one seventh of its parameters. In future work, we will focus on extending the PHANTOM framework's applicability. A separate context encoder, similar to the background estimator, whose RGB features are fused with the spectral features in the latent space, would let the spectral signal drive detection while visual context only confirms or rejects candidates. A second direction is an advection–diffusion transport prior that acts on the predicted map itself, with advection driven by the wind and diffusion set by the local turbulence, tested on controlled-release experiments in which the source location and the wind measured at the site are known [47]; the plume masks of PHANTOM can also serve wind-based emission-rate estimation [45]. We plan to adapt and validate its performance on data from other airborne and satellite sensors, such as TROPOMI, to enable broader environmental monitoring. Furthermore, the methodology can be generalized to detect other greenhouse gases or different target materials with distinct spectral signatures, opening new avenues for physics-informed remote sensing applications.

INFORMATION ON THE REPRODUCIBILITY OF RESULTS

The dataset can be directly downloaded from the provided link [12]. These results can be easily reproduced using the trained model, which is available on Kaggle at https://www.kaggle.com/models/sachingiri348/phantom. The visualization code for the trained model can be accessed on Google Colab: https://colab.research.google.com/drive/1COOFV7Re-6FLPMktRTrAA9vpjmy2Q05V?usp=sharing

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